Chapter 9: Contact-Borne Diseases

Intended Learning Objectives

By the end of this chapter, students should be able to:

1. **Define contact transmission and distinguish between direct and indirect contact.**
2. **Describe the epidemiology, clinical features, and prevention of key contact-borne diseases.**
3. **Explain the pathogenesis and control strategies for rabies and tetanus.**
4. **Understand vaccination schedules for rabies and tetanus, including protocols during pregnancy.**
5. **Apply public health measures to reduce transmission in healthcare and community settings.**

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1. Introduction

Contact-borne diseases are transmitted through physical contact with infected individuals, contaminated surfaces, or animal bites. Transmission may be:

- **Direct contact: skin-to-skin, mucous membranes, bites, scratches**
- **Indirect contact: via contaminated objects (fomites), such as towels, instruments, or soil**

These diseases range from mild skin infections to life-threatening neuroinvasive conditions. Rabies and tetanus are two critical contact-transmitted diseases with high fatality rates if untreated, yet both are vaccine-preventable.

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2. Overview of Transmission

Type	Mechanism	Examples
Direct contact	Skin, mucosa, bites, scratches	Rabies, scabies, impetigo
Indirect contact	Contaminated surfaces, soil, instruments	Tetanus, MRSA, ringworm

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3. Key Contact-Borne Diseases

3.1 Rabies

Causative Agent:
Rabies virus (*Lyssavirus*, Rhabdoviridae family)

Reservoirs:
Dogs (primary in Egypt), bats, foxes, raccoons

Mode of Transmission:

- **Bite or scratch from infected animal**

- **Saliva contact with broken skin or mucosa**

Incubation Period:
Typically 1–3 months (can range from days to years depending on bite site and viral load)

Clinical Features:

- **Prodromal phase: Fever, malaise, pain at bite site**
- **Neurological phase: Hydrophobia, agitation, hallucinations, paralysis**
- **Coma and death: Once symptoms appear, rabies is almost always fatal**

Diagnosis:

- **Clinical suspicion based on exposure**
- **RT-PCR from saliva, skin biopsy, or CSF**
- **Postmortem confirmation from brain tissue**

Prevention and Control:

- **Pre-exposure prophylaxis (PrEP): For high-risk groups (veterinarians, lab workers)**
- **Post-exposure prophylaxis (PEP): Immediate wound cleaning + vaccine ± rabies immunoglobulin (RIG)**

🧪 Rabies Vaccination Schedule

🔹 Pre-Exposure Prophylaxis (PrEP)

- **Indicated for: High-risk occupations**

- **Schedule:**
  - **Day 0, Day 7, Day 21 or 28 (intramuscular or intradermal)**
  - **Booster every 3–5 years if risk persists**

🔹 **Post-Exposure Prophylaxis (PEP)**

- **For previously unvaccinated individuals:**
  - **Rabies vaccine: 4 doses on Days 0, 3, 7, and 14**
  - **Rabies Immunoglobulin (RIG): Infiltrated around wound on Day 0**
- **For previously vaccinated individuals:**
  - **Rabies vaccine only: 2 doses on Days 0 and 3**
  - **No RIG needed**

Public Health Notes:

- **Rabies is a notifiable disease**
- **Dog vaccination campaigns are essential**
- **Education on wound care and prompt reporting is critical**

3.2 Tetanus

Causative Agent:

Clostridium tetani **– anaerobic, spore-forming bacterium**

Reservoir:

Soil, dust, animal feces

Mode of Transmission:

- **Entry of spores into wounds, burns, or punctures**
- **Non-communicable: not spread person-to-person**

Incubation Period:

3–21 days (average 7 days)

Clinical Features:

- **Early signs: Jaw stiffness (trismus), neck rigidity**
- **Progression: Muscle spasms, opisthotonos, respiratory failure**
- **Neonatal tetanus: Infection via contaminated umbilical stump**

Diagnosis:

- **Clinical presentation**
- **No specific laboratory test; diagnosis is clinical**

Prevention and Control:

- **Vaccination is the cornerstone**
- **Proper wound care and hygiene**
- **Tetanus toxoid boosters for injuries**

🧪 Tetanus Vaccination Schedule

🔹 Routine Childhood Immunization (Egypt EPI)

- **DTaP vaccine at:**

- **2 months**
- **4 months**
- **6 months**
- **Booster at 18 months**
- **Booster at 4–6 years**
- **Tdap booster at 11–12 years**

◆ **Adult Boosters**

- **Td vaccine every 10 years**
- **Tdap once in adulthood if not previously received**

◆ **Post-Exposure Prophylaxis**

Wound Type	**Vaccination History**	**Recommended Action**
Clean minor	**≥3 doses**	**No vaccine needed**
Dirty wound	**≥3 doses**	**Td booster if >5 years**
Any wound	**<3 doses or unknown**	**Td + TIG (Tetanus Immunoglobulin)**

Tetanus Vaccination in Pregnancy

Goal: Prevent maternal and neonatal tetanus

Recommended Schedule (WHO & Egyptian MOH):

Dose	Timing	Protection
TT1	After first trimester	No protection yet
TT2	≥4 weeks after TT1	3 years protection
TT3	≥6 months after TT2	5 years protection
TT4	≥1 year after TT3	10 years protection
TT5	≥1 year after TT4	Lifetime protection

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4. Other Contact-Borne Diseases (Brief Overview)

Disease	Agent	Transmission	Prevention
Impetigo	*S. aureus*, *S. pyogenes*	Skin contact	Hygiene, topical antibiotics
Scabies	*Sarcoptes scabiei*	Prolonged skin contact	Treat contacts, wash linens
Ringworm	Dermatophytes	Skin/fomite contact	Antifungals, hygiene
MRSA	Resistant *S. aureus*	Direct/indirect contact	Contact precautions

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Multiple Choice Questions (MCQs)

1. Which of the following diseases is non-communicable and acquired through contaminated wounds?

A. Rabies

B. Tetanus

C. Scabies

D. MRSA

Correct Answer: B. Tetanus

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2. Rabies post-exposure prophylaxis includes:

A. Antibiotics and tetanus vaccine

B. Rabies vaccine and immunoglobulin

C. Antifungal cream and wound dressing

D. Isolation and antiviral tablets

Correct Answer: B. Rabies vaccine and immunoglobulin

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3. How often should adults receive a tetanus booster if fully vaccinated?

A. Every 5 years

B. Every 10 years

C. Once in a lifetime

D. Every year

Correct Answer: B. Every 10 years

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4. Which vaccine is recommended during pregnancy to prevent neonatal tetanus?

A. DTaP

B. Rabies vaccine

C. TT (Tetanus Toxoid)

D. Hepatitis B vaccine

Correct Answer: C. TT (Tetanus Toxoid)

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5. Which of the following statements about rabies is TRUE?

A. It is curable after symptoms appear

B. It is transmitted through contaminated water

C. It is preventable by vaccination and wound care

D. It is a fungal disease

Correct Answer: C. It is preventable by vaccination and wound care